RETINA-4FN Basic Viewer User Guide (ROS2)

Smart Radar System, Inc.

SMART RADAR SYSTEM

Environment Setup (on Linux)

Source Code Build

Running RETINA-4FN with RVIZ

Environment Setup
(on Linux)

Environment settings

1. Setup Linux (Ubuntu)

   Ubuntu 22.04 LTS would be highly recommended

   Latest ROS2 version is Iron

   ※ Ubuntu version check

   1) ctrl + alt + t : terminal open

   2) lsb_release –a

```
newton@Fermat:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.3 LTS
Release:        22.04
Codename:       jammy
```

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Environment settings

2. Install ROS2

Ubuntu 22.04 LTS → ROS2 “Iron”

→ Iron Installation - ROS2 Documentation

※ Ubuntu version & ROS2 version MUST be matched

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Environment settings

2-1) Install ROS2 “**Iron**” on **Ubuntu 22.04**

- Setup Sources

```
$ sudo apt install software-properties-common
$ sudo add-apt-repository universe
$ sudo apt update && sudo apt install curl -y
$ sudo curl -sSL https://raw.githubusercontent.com/ros/rosdistro/master/ros.key -o /usr/share/keyrings/ros-archive-keyring.gpg
$ echo "deb [arch=$(dpkg --print-architecture) signed-by=/usr/share/keyrings/ros-archive-keyring.gpg] http://packages.ros.org/ros2/ubuntu $(. /etc/os-release && echo $UBUNTU_CODENAME) main" | sudo tee /etc/apt/sources.list.d/ros2.list > /dev/null
$ sudo apt update && sudo apt install ros-dev-tools
```

- Install ROS 2 packages

```
$ sudo apt update
$ sudo apt upgrade
$ sudo apt install ros-iron-desktop
```

※ Install Line by Line in ‘Terminal’

- Sourcing the setup script

```
$ echo "source /opt/ros/iron/setup.bash" >> ~/.bashrc
```

Environment settings

3. Make directory environment

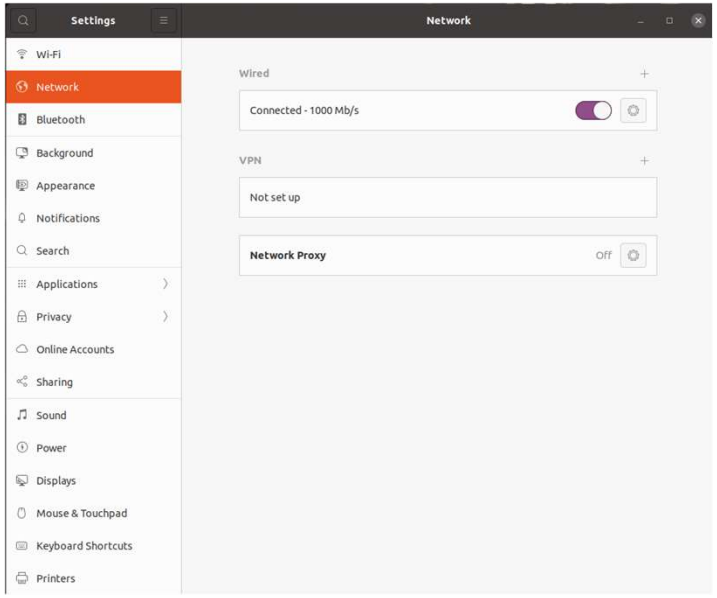
- Copy or download the file: 4fn_basic_viewer_ros2.tgz
- Check the file path
- Open the terminal and make working directory you want, like dev_ws
- Input the below command to unzip the files in the working directory

1) tar xfz [file path]/4fn_basic_viewer_ros2.tgz

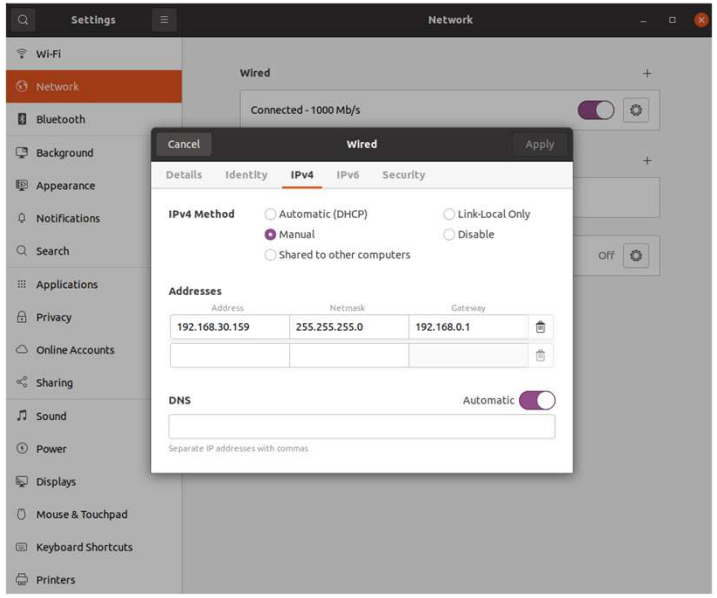
ex) If you copy 4fn_basic_viewer_ros2.tgz in ‘Download’ directory → tar xfz Download/4fn_basic_viewer_ros2.tgz

→ After a successful unzip, you could see the ‘4fn_basic_viewer’ folder on your working directory

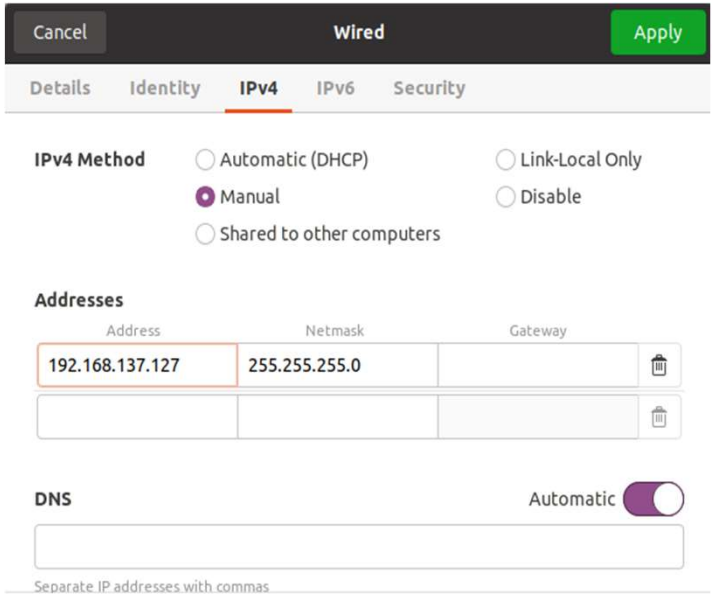
Static IP setup



1. Open ***Network Settings***



2. Select the ***Internet Protocol Version 4 (IPv4) from Wired Network***



3. Set any static IP which will not duplicated with the IP of RETINA-4FN in ***192.168.137.xxx***

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Source Code Build

Build Steps

1. Open the Terminal (***ctrl +alt + t)***
2. Go to your working directory
3. Input the below command on the Terminal to build the source code

```
$ colcon build --packages-select 4fn_basic_viewer
```

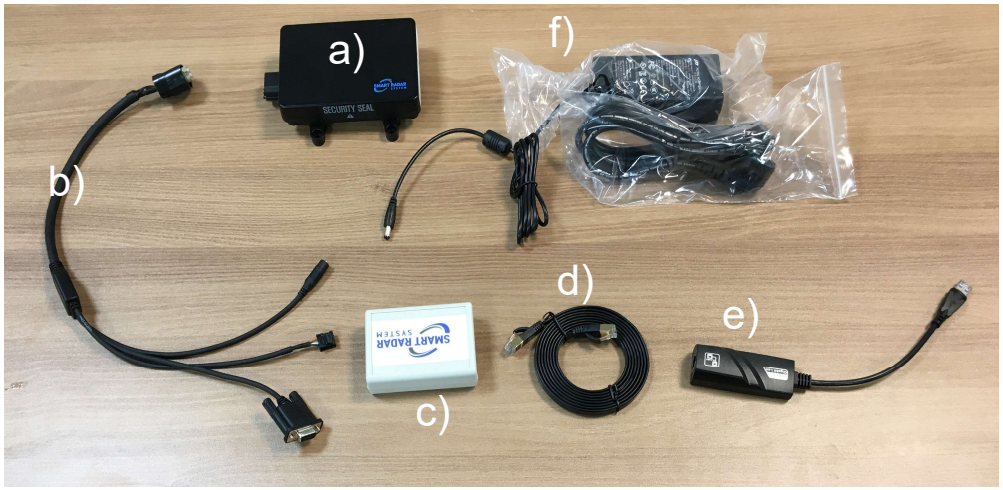
```
newton@Fermat:~/work/ros2/dev_ws$ colcon build --packages-select 4fn_basic_viewer
WARNING: Package name "4fn_basic_viewer" does not follow the naming conventions. It should start with a lower case letter and only cont
ain lower case letters, digits, underscores, and dashes.
Starting >>> 4fn_basic_viewer
Finished <<< 4fn_basic_viewer [0.36s]

Summary: 1 package finished [0.96s]
newton@Fermat:~/work/ros2/dev_ws$
```

Running RETINA-4FN with RVIZ

Components

1. RETINA-4FN consists of:

a) Retina 4FN Radar
b) Retina 4FN Connector Cable
c) Media Converter
d) Ethernet cable
e) USB3.0 Gigabit Lan Adapter (NOT included in the package, Recommended for 1Gb connection)
f) 12V Adapter
g) PC



All components are connected

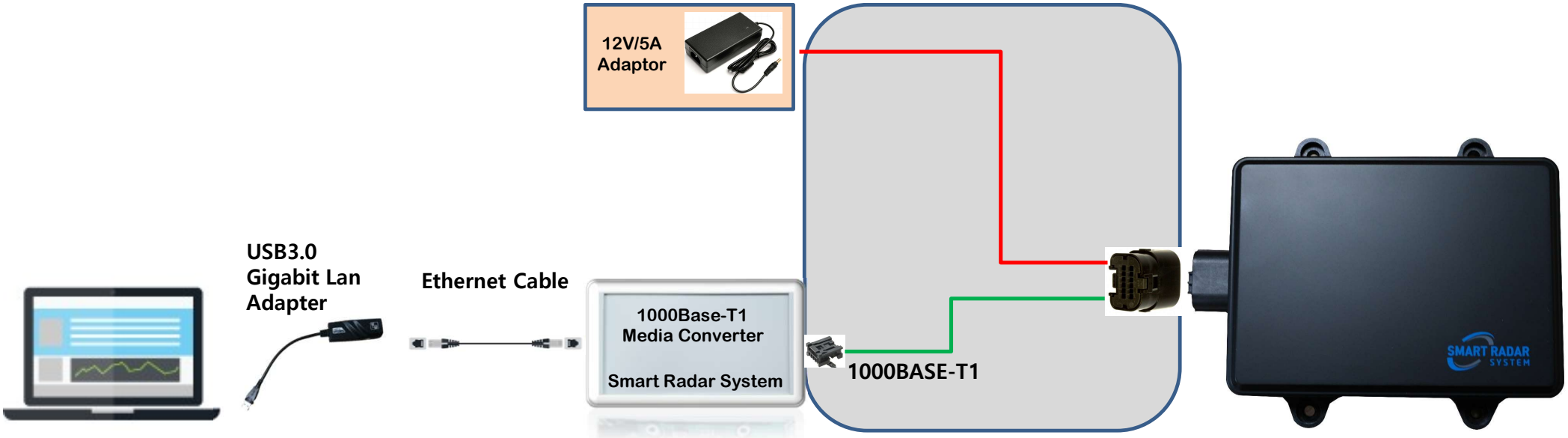
HW Setup

1. The internal test setting of the RETINA-4FN consists of:

- RETINA-4FN Radar.
- RETINA-4FN Connector Cable.
- 12V/5A DC Adapter.
- 1000Base-T1 Media Converter.
- Ethernet Cable
- USB3.0 Gigabit Lan Adapter

2. Interface type and features

- 1000Base-T1 : Point cloud output / Firmware download



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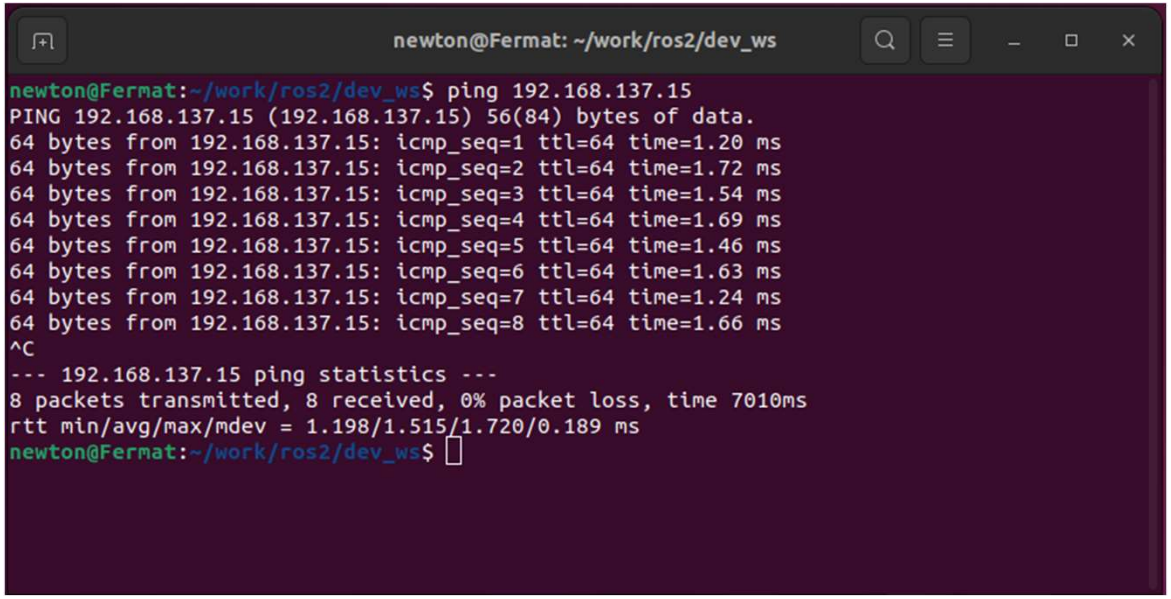
Check the Network Connection

1. Run the ***Terminal Program (ctrl +alt + t)***
2. Run the following command on the Terminal

```
$ ping 192.168.137.15
```



The default IP address of RETINA-4FN is ***192.168.137.15***
If ping does not work, please check the static IP address setting of the PC Ethernet port.

Configuration

❖ Edit the IP address of RETINA-4FN in the yaml file

```
install/4fn_basic_viewer/share/4fn_basic_viewer/config/front_center.yaml
```

If you need to update the config file, please don't edit the file with ***Windows notepad***.
You need to use any editor on Linux or source code editors on Windows such as ***Visual Studio Code***.
Otherwise, you will have unexpected errors with new line character mismatch between Linux and Windows.

Before using the package, please modify the correct IP address of radar on config file to connect without problem.
The default IP address of RETINA-4FN would be 192.168.137.15
z_offset means height from the ground and the unit is cm.

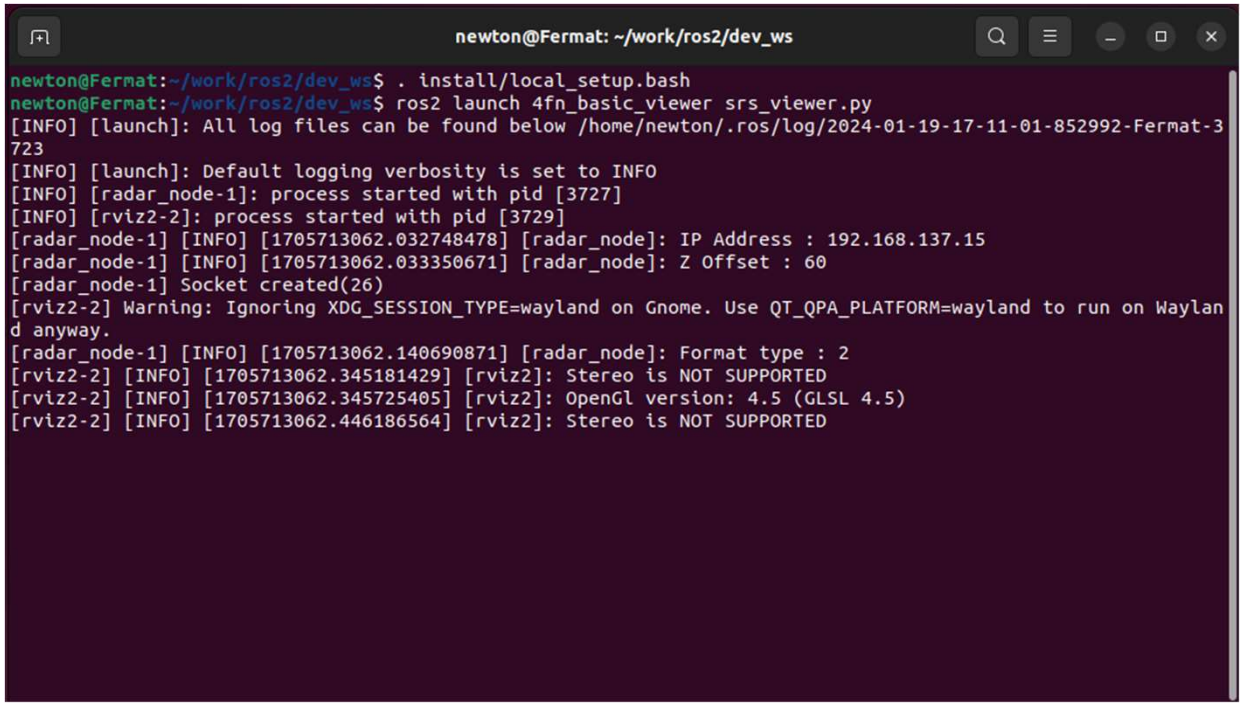
```
radar_node:
  ros__parameters:

    position:     0
    ip:           "192.168.137.15"
    z_offset:     60
```

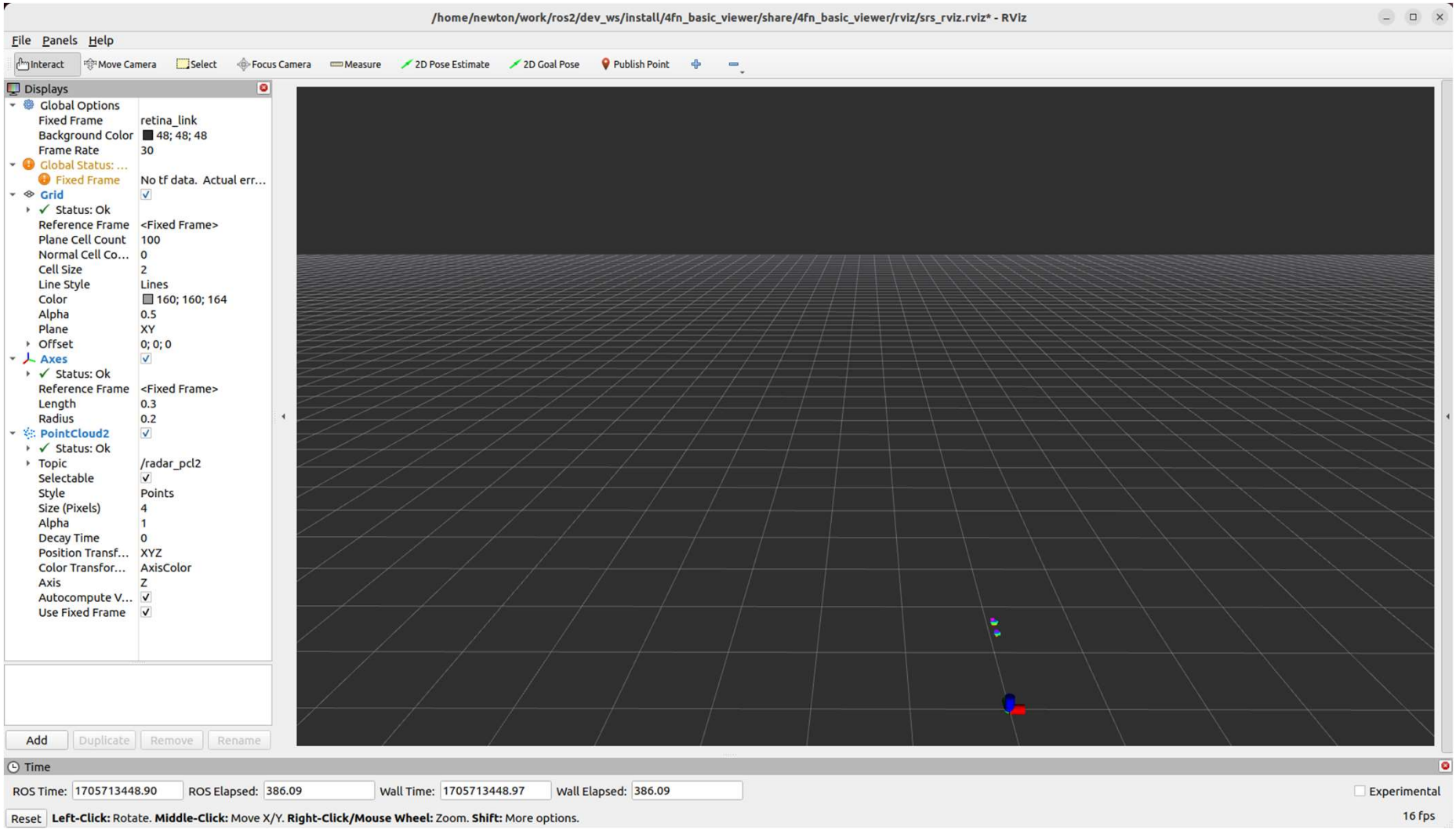
Commands on Terminal To Display the Points from Radar

1. Run the Terminal Program (***ctrl +alt + t)***
2. Go to your working directory
3. Input the below command on the Terminal

```
$ . install/local_setup.bash
$ ros2 launch 4fn_basic_viewer srs_viewer.py
```



RVIZ Display



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